

REMARKS ON SIX-TANGENTIAL CONICS TO SMOOTH PLANE SEXTICS

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ABSTRACT.

A smooth plane sextic together with a configuration of splitting conics carries geometric information that is not captured by the incidence data of the curves in the plane. This information becomes visible on the double cover of $\mathbb{P}^2$ branched along the sextic: each splitting conic lifts to two components, and the intersection numbers between chosen components define a weighted graph, considered up to switching and relabeling. These graphs, together with the lattices generated by the lifted components and the pullback of a line, provide finer invariants of the configuration. Our aim is to investigate the abundance of configurations that share the same plane incidence data but are distinguished by these invariants, and to give explicit computational classifications for configurations with a small number of conics. (By Codex)

Let k be a positive integer. We consider the complete graph with vertex set $V_k = \{1, \dots, k\}$ and edge set

$$E_k := \{[i, j] \mid i, j \in V_k, i < j\}.$$

Let m be a positive integer, and we put

$$M := \{-m, -m+1, \dots, m\}.$$

Let $\mathcal{W}_k$ be the set of weights $w: E_k \rightarrow M$, that is, $\mathcal{W}_k$ is the set of weighted complete graphs on V_k with weights in M . We let the symmetric group $\mathfrak{S}_k$ act on V_k from the right. Then $\mathfrak{S}_k$ acts on $\mathcal{W}_k$ from the left. We define Sw_k to be the multiplicative group of all mappings $s: V_k \rightarrow \{\pm 1\}$, which acts on $\mathcal{W}_k$ by switching, that is,

$$(sw)([i, j]) = s(i)s(j)w([i, j])$$

for $s \in \text{Sw}_k$ and $[i, j] \in E_k$. Then the semidirect product

$$G_k := \text{Sw}_k \rtimes \mathfrak{S}_k$$

acts on $\mathcal{W}_k$ from the left. The purpose of this section is to describe an efficient algorithm for obtaining the list of orbits of the action of G_k on $\mathcal{W}_k$.

First, we introduce a total order on the set E_k of edges by

$$[i, j] < [i', j'] \iff j < j' \text{ or } (j = j' \text{ and } i < i').$$

Then we express elements w of $\mathcal{W}_k$ by arranging the values $w([i, j])$ in this order:

$$w = (w([1, 2]), w([1, 3]), w([2, 3]), w([1, 4]), \dots).$$

Let $\text{ab}(w)$ denote the vector obtained from w by taking the absolute value of each component. We put $\text{ab}(M) := \{0, 1, \dots, m\}$. Then we have a mapping

$$(0.1) \quad \text{ab}(\cdot) : \mathcal{W}_k \longrightarrow \text{ab}(\mathcal{W}_k) := (\text{ab}(M))^{E_k}, \quad w \longmapsto \text{ab}(w).$$

Definition 0.1. Suppose that a group G acts on a set X , and that X is equipped with a total order $<_X$. An element $x \in X$ is $(G, <_X)$ -*minimal* if x is minimal with respect to $<_X$ in the orbit $G \cdot x$.

Notice that $\mathfrak{S}_k$ acts on $\text{ab}(\mathcal{W}_k)$, and the mapping (0.1) is $\mathfrak{S}_k$ -equivariant. Moreover, the action of Sw_k on $\mathcal{W}_k$ preserves each fiber of (0.1). We then define a total order $<_{\mathcal{W}}$ on $\mathcal{W}_k$ by

$$w <_{\mathcal{W}} w' \iff \text{ab}(w) <_{\text{lex}} \text{ab}(w') \quad \text{or} \quad (\text{ab}(w) = \text{ab}(w') \quad \text{and} \quad w <_{\text{lex}} w').$$

We obtain the list of G_k -orbits in $\mathcal{W}_k$ by enumerating all $(G_k, <_{\mathcal{W}})$ -minimal elements in $\mathcal{W}_k$.

We define the notion of G_k -*quasi-minimality* by the following:

$$w \text{ is } G_k\text{-quasi-minimal} \iff \begin{array}{l} \text{ab}(w) \text{ is } (\mathfrak{S}_k, <_{\text{lex}})\text{-minimal, and} \\ w \text{ is } (\text{Sw}_k, <_{\text{lex}})\text{-minimal in } \text{ab}^{-1}(\text{ab}(w)). \end{array}$$

It is obvious that, if $w \in \mathcal{W}_k$ is $(G_k, <_{\mathcal{W}})$ -minimal then $w \in \mathcal{W}_k$ is G_k -quasi-minimal while the opposite implication does not necessarily hold. We first make the list of G_k -quasi-minimal elements, and then we choose from them $(G_k, <_{\mathcal{W}})$ -minimal elements.

To make the list of G_k -quasi-minimal elements, we use the induction on k . Suppose that $k' < k$. By taking the prefix of length $k'(k' - 1)/2$ of each $u \in \text{ab}(\mathcal{W}_k)$, we obtain a mapping

$$\text{ab}(\mathcal{W}_k) \longrightarrow \text{ab}(\mathcal{W}_{k'}), \quad u \longmapsto u|_{k'}.$$

It is obvious that, if $u \in \text{ab}(\mathcal{W}_k)$ is $(\mathfrak{S}_k, <_{\text{lex}})$ -minimal, then $u|_{k'}$ must be $(\mathfrak{S}_{k'}, <_{\text{lex}})$ -minimal.

For an $(\mathfrak{S}_k, <_{\text{lex}})$ -minimal element $u \in \text{ab}(\mathcal{W}_k)$,

$$\text{ab}^{-1}(u) \longrightarrow \text{ab}^{-1}(u|_{k'}), \quad w \longmapsto w|_{k'}.$$

It is obvious that, if $w \in \text{ab}^{-1}(u)$ is $(\text{Sw}_k, <_{\text{lex}})$ -minimal, then $w|_{k'}$ must be $(\text{Sw}_{k'}, <_{\text{lex}})$ -minimal. More strongly, $w \in \text{ab}^{-1}(u)$ is $(\text{Sw}_k, <_{\text{lex}})$ -minimal only if $w|_{k'}$ is $(\text{Sw}_{k'}, <_{\text{lex}})$ -minimal and the first non-zero element of the suffix $w - w|_{k'}$ is negative. (The opposite implication does not necessarily hold.) We say that w is $(\mathfrak{S}_k, <_{\text{lex}})$ -quasi-minimal if, for any $k' < k$, $w|_{k'}$ is $(\mathfrak{S}_{k'}, <_{\text{lex}})$ -quasi-minimal and the first non-zero element of the suffix $w - w|_{k'}$ is negative.

We enumerate all $(\mathfrak{S}_k, <_{\text{lex}})$ -minimal elements and then, for each $(\mathfrak{S}_k, <_{\text{lex}})$ -minimal element $u \in \text{ab}(\mathcal{W}_k)$, we enumerate all $(\text{Sw}_k, <_{\text{lex}})$ -quasi-minimal elements in $\text{ab}^{-1}(u)$.

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